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Research Article

Machine Learning-Guided Design of High-Entropy FeCoCrMnCu Layered Double Hydroxides for Efficient Oxygen Evolution in Alkaline Media

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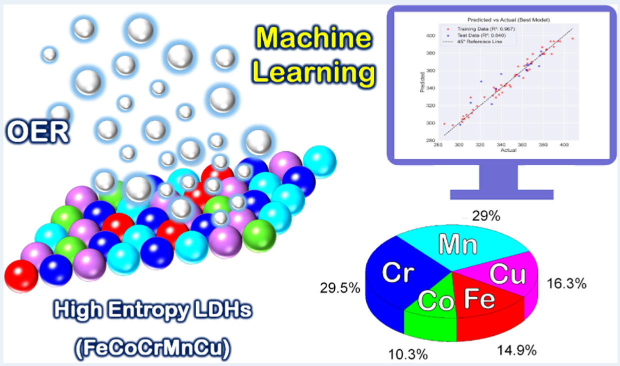
ABSTRACT: The discovery of high-performance oxygen evolution reaction (OER) catalysts is often hindered by the vast compositional space of high-entropy materials, making conventional trial-and-error methods time-consuming and resource-intensive. In this work, we demonstrate a machine learning (ML)-guided strategy for the design of high-entropy FeCoCrMnCu layered double hydroxides (LDHs) as advanced OER catalysts in alkaline media. An experimental data set of only 70 compositions was used to train an Extreme Gradient Boosting (XGBoost) regression model, which achieved high predictive accuracy (R^2 = 0.84, RMSE = 9.95 mV). The ML model identified an optimal composition ($Fe_{0.15}Co_{0.10}Cr_{0.30}Mn_{0.30}Cu_{0.15}$) with a predicted overpotential of 261.3 mV, closely matching the experimentally obtained 270 mV (error ∼ 3%). This approach effectively reduced the need for exhaustive testing of more than 10,626 possible compositions, achieving a 99.3% reduction in time and effort. The ML-optimized catalyst exhibited favorable morphology, homogeneous elemental distribution, and strong intrinsic activity, with a Tafel slope of 74.2 mV dec^{-1}, high turnover frequency (0.225 s^{-1}), and stable operation for 72 h. This study highlights the power of integrating ML with entropy-driven materials design to accelerate the development of next-generation electrocatalysts.



KEYWORDS: *machine learning optimization, high entropy LDHs, OER, data-driven materials discovery, XGBoost regression, catalyst composition engineering*

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■ INTRODUCTION

The transition toward a carbon-neutral future demands scalable and sustainable technologies to produce clean fuels.[1−3] Among them, electrochemical water splitting has emerged as a promising and environmentally benign strategy for generating high-purity hydrogen, an ideal energy carrier owing to its high energy density and zero carbon emissions upon use.[4] Water electrolysis involves two half-reactions: the hydrogen evolution reaction (HER) at the cathode and the oxygen evolution reaction (OER) at the anode.[5,6] The OER is a multistep process that involves the concerted transfer of four electrons and four protons, resulting in the formation of molecular oxygen (O_2).[7] Although the thermodynamic potential required for OER is 1.23 V vs SHE (standard hydrogen electrode), in practice, a much higher potential typically between 1.8 and 2.0 V vs SHE is required to overcome the intrinsic activation barriers.[8,9] Noble metals such as Ir, Ru, Pt, and Au have demonstrated outstanding activity in reducing these overpotentials, with IrO_2 and RuO_2 being the most effective OER electrocatalysts. However, their high cost, limited supply, and poor long-term stability under harsh alkaline or acidic conditions hinder their widespread deployment.[10−12]

Mechanistically, OER proceeds via multiple catalytic steps, including the adsorption of hydroxide ions (OH^-), the formation of high-energy intermediates (*O and *OOH), and the eventual desorption of O_2. These steps involve high-energy transitions that significantly slow down the reaction kinetics.[13−15] As a result, overcoming these kinetic limitations is critical to improve the overall efficiency of water splitting and other electrochemical energy conversion technologies such as metal−air batteries and regenerative fuel cells. The development of cost-effective, earth-abundant, and intrinsically active

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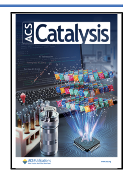

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electrocatalysts is therefore essential to reduce energy losses, accelerate reaction rates, and enable practical hydrogen production.[16]

In this context, layered double hydroxides (LDHs) have emerged as a promising class of OER catalysts due to their tunable composition, high surface area, and abundant metal active sites. LDHs have a general formula of $[M_{1-x}^{2+}M_x^{3+}(OH)_2]^{x+}(A^{n-})^{x/n}\cdot mH_2O$, where divalent ($M^{2+}$) and trivalent ($M^{3+}$) metal cations form positively charged layers, balanced by interlayer anions and water molecules.[17] These materials feature edge-sharing octahedra arranged in stacked two-dimensional (2D) layers. Key features that make LDHs ideal for OER include their flexible metal cation composition, the tunability of interlayer anions to modulate catalytic behavior, and the ability to exfoliate into ultrathin nanosheets, thereby significantly enhancing electrochemical surface area and reaction kinetics.[18,19] Recently, high-entropy materials (HEMs) defined by the inclusion of five or more principal elements in near-equimolar ratios have attracted increasing attention in catalysis. The high configurational entropy in these systems stabilizes single-phase structures and introduces synergistic interactions between multiple transition metal centers, which can modulate electronic properties, promote oxygen vacancy formation, and enhance intrinsic catalytic activity.[20−22]

Recent advances in high-entropy layered double hydroxides (HE-LDHs) have demonstrated their strong potential as next-generation electrocatalysts for OER.[23] For instance, Wang et al. reported a MnFeCoNiCu LDHs decorated with gold single atoms and oxygen vacancies (AuSA-MnFeCoNiCu LDHs), which exhibited a remarkably low overpotential of 213 mV at 10 mA cm^{-2} and a high mass activity of 732.925 A g^{-1} at 250 mV overpotential in 1.0 M KOH, with excellent long-term stability over 700 h at ∼100 mA cm^{-2}.[24] In another study, Cao et al. synthesized a CuO@FeCoNiCrMn LDHs heterostructure via in situ electrochemical deposition. The catalyst achieved an overpotential of 251 mV at 100 mA cm^{-2}, with a Tafel slope of 34.1 mV dec^{-1} and showed sustained durability for over 40 h. Their CuO@FeCoNiCrMn LDH‖Pt/C electrolyzer achieved current densities of 10 and 100 mA cm^{-2} at low voltages of 1.50 and 1.54 V, respectively.[25] Additionally, Wondimu et al. developed a FeAlNiCoMn HE-LDHs supported on carbon nanotubes (FeAlNiCoMn-HE-LDHs/CNT), achieving an overpotential of 243 mV at 10 mA cm^{-2} and a Tafel slope of 46.54 mV dec^{-1}, outperforming even IrO_2 benchmarks. Impressively, the catalyst demonstrated outstanding durability, with only ∼1.5% drop in current density after 250 h at 150 mA cm^{-2} and slight improvement at lower current densities.[26] Within our own research group, a high-entropy FCCMZ LDHs (Fe, Co, Cr, Mn, Zn) was synthesized via a simple hydrothermal method, delivering excellent activity toward both OER and urea oxidation reaction (UOR). The optimized FCCMZ LDHs exhibited an OER overpotential of 185 mV at 10 mA cm^{-2} with a Tafel slope of 49.7 mV dec^{-1}, and a low UOR potential of 250 mV vs Hg/HgO under alkaline conditions.[27] Despite their promise, a major challenge in HEM catalyst development lies in the virtually limitless compositional space, making experimental discovery time-consuming and inefficient. Machine learning (ML), as an effective means of implementing artificial intelligence,[28−32] provides a powerful solution to this problem by enabling rapid screening and optimization of material compositions based on known structure, property, and performance relationships. By training predictive models on curated data sets of catalyst compositions and electrochemical performance, it is possible to identify optimal combinations without exhaustive experimental efforts.[33−35]

In our previous work, we have synthesized FeCoCrMnCu-LDHs (0.45 mM each) and analyzed its OER activity (295 mV at 10 mA cm^{-2}) without using ML model.[36] But in this work, we report a machine learning-guided discovery and synthesis of a high-entropy FeCoCrMnCu-LDHs electrocatalyst for the OER in alkaline media. Extreme Gradient Boosting (XGBoost) regression model was trained using physical descriptors of transition metals and performance data from initial data set.[37] The model predicted that a composition of $Fe_{0.15}Co_{0.10}Cr_{0.30}Mn_{0.30}Cu_{0.15}$ would exhibit optimal OER activity. This predicted LDHs catalyst was synthesized via a facile hydrothermal method, and its performance was validated using a suite of structural, chemical, and electrochemical techniques.

The synthesized catalyst exhibited a low overpotential of 270 mV at 10 mA cm^{-2}, a Tafel slope of 74.2 mV dec^{-1}, and excellent long-term stability over 72 h. The high surface area (233.5 m^2 g^{-1}), abundant oxygen vacancies, and synergistic multimetal composition contributed to its superior performance. Although several high-entropy LDHs-based catalysts reported in the literature exhibit lower overpotentials or Tafel slopes,[24−26] many rely on noble-metal decoration, conductive supports, or extensive experimental optimization. In contrast, the present work focuses on validating a data-efficient machine learning framework capable of accurately predicting optimal compositions using a minimal experimental data set. The close agreement between predicted and experimental overpotentials (∼3% error) highlights the robustness of the ML-guided approach and underscores its potential for accelerating catalyst discovery beyond trial-and-error experimentation. This study illustrates a successful integration of ML and entropy-driven material design, demonstrating an effective pathway for the accelerated development of next-generation electrocatalysts.

■ EXPERIMENTAL SECTION

Materials and Synthesis Procedure

The machine learning-optimized FeCoCrMnCu (FCCMC) LDHs catalyst was synthesized via a facile one-step hydrothermal method. $Fe(NO_3)_3\cdot 9H_2O$, $Co(NO_3)_2\cdot 6H_2O$, $Cr(NO_3)_3\cdot 9H_2O$, $Mn(NO_3)_2\cdot 6H_2O$, and $Cu(NO_3)_2\cdot 6H_2O$ were accurately weighed based on the ML-predicted composition molar ratio (Fe/Co/Cr/Mn/Cu = 0.15:0.10:0.30:0.30:0.15). These precursors were dissolved in 40 mL of deionized water containing 10 mmol of urea. The solution was magnetically stirred for 1 h to ensure complete dissolution and homogeneity. Subsequently, the pH of the solution was adjusted to approximately 10 using 1 M NaOH and 1 M Na_2CO_3. The resulting mixture was transferred into a 100 mL Teflon-lined stainless-steel autoclave and subjected to hydrothermal treatment at 180 °C for 12 h. After naturally cooling to room temperature, the solid product was collected by centrifugation, thoroughly washed several times with deionized water and ethanol, and then dried in a vacuum oven at 60 °C for 12 h. The obtained powder was labeled as FCCMC LDHs, representing the high-entropy, ML-optimized FeCoCrMnCu LDHs.

Characterization Techniques

The morphology and structure of FCCMC LDHs were analyzed using field-emission scanning electron microscopy (FE-SEM) and high-resolution transmission electron microscopy (HR-TEM), with energy-dispersive X-ray spectroscopy (EDX) for elemental mapping. SEM (ZEISS ULTRA PLUS) was used to observe surface features and nanosheet morphology, while HR-TEM (JEOL JEM-2010, 200

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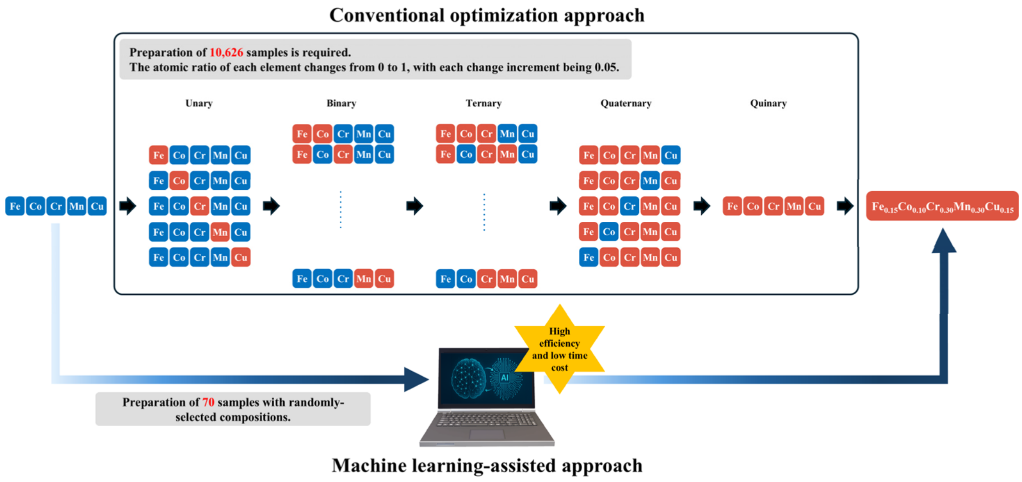

Figure 1. Schematic illustration of conventional optimization vs ML-guided design of high-entropy LDHs.

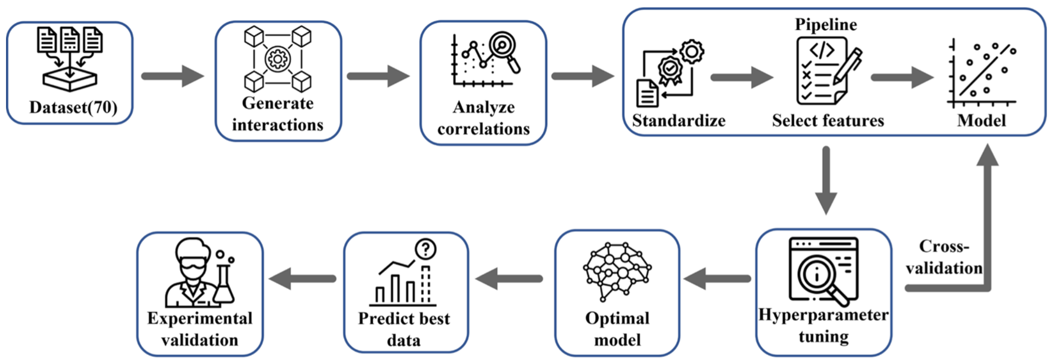

Figure 2. Workflow of machine learning optimization.

kV) provided detailed lattice images. Crystallographic analysis was performed via X-ray diffraction (XRD) using a Bruker D8 SSS diffractometer with Cu Kα radiation (λ = 1.54 Å). X-ray photoelectron spectroscopy (XPS, ULVAC-PHI PHI 5000 VersaProbe, Al Kα) was employed to determine the electronic states and oxidation levels of elements. Nitrogen adsorption−desorption measurements at 77.4 K (Micromeritics ASAP 2020) were used to calculate the specific surface area using the Brunauer−Emmett−Teller (BET) method, providing insights into the material's porosity and surface accessibility. The electrocatalytic performance of FCCMC LDHs was assessed using linear sweep voltammetry (LSV), Tafel analysis, chronopotentiometry, electrochemical impedance spectroscopy (EIS), and electrochemically active surface area (ECSA) estimation. All tests were performed on a CHI6279E electrochemical workstation in a standard three-electrode setup for OER, employing Ni foam coated with LDHs as the working electrode, Hg/HgO as the reference, and Pt wire as the counter electrode. All the potentials were converted to the reversible hydrogen electrode (RHE) scale.

Experimental Data Set Used for Machine Learning

In this study, a data set comprising 70 distinct high-entropy layered double hydroxides (HE LDHs) compositions was generated by systematically varying the molar ratios of five transition metals: Fe, Co, Cr, Mn, and Cu (Table S1). Each composition was normalized such that the sum of all elemental molar fractions equaled one, ensuring uniformity and comparability across the data set. Figure S1 illustrates the distribution ranges of the five constituent elements, confirming that the data set captures a wide diversity of elemental combinations. This compositional variability enhances the data set's representativeness and provides a solid foundation for developing a robust machine learning model. The measured overpotential values at 10 mA cm^{-2} from the LSV curves spanned from 286 mV to 408 mV (Figure S2), with an average of approximately 350 mV and a standard deviation of 29 mV, as shown in Figure S3. The distribution of overpotential exhibited a slight right skew, indicating a higher occurrence of relatively less efficient compositions, which is typical in broad exploratory data sets.

Results and Discussion

The schematic illustration (Figure 1) highlights the significant advantage of integrating machine learning into catalyst discovery compared to conventional optimization approaches. Traditionally, the development of high-entropy materials requires the preparation and testing of a vast number of samples, beginning with unary and binary alloys, and progressively moving through ternary, quaternary, and quinary systems. Such exhaustive experimental exploration not only demands considerable time, cost, and resources but also limits the efficiency of identifying the most active catalytic composition. In contrast, the machine learning-guided strategy dramatically reduces the experimental workload by requiring only a limited number of initial training samples. By leveraging predictive models, the compositional space can be rapidly screened, and the most promising candidates can be identified with high accuracy and efficiency. In this study, ML effectively predicted the optimal $Fe_{0.15}Co_{0.10}Cr_{0.30}Mn_{0.30}Cu_{0.15}$ LDHs composition, achieving strong agreement with experimental validation, thereby underscoring the transformative role of computational intelligence in accelerating the design of high-entropy LDHs for OER.

Generation of the Training Data Set by Experiments

To develop a ML model for the rational design of high-entropy electrocatalysts, a training data set was constructed using experimentally synthesized FeCoCrMnCu (FCCMC) LDHs. The primary objective was to correlate catalyst composition with OER performance, specifically the overpotential required to achieve a current density of 10 mA cm^{-2} in an alkaline medium. The key input

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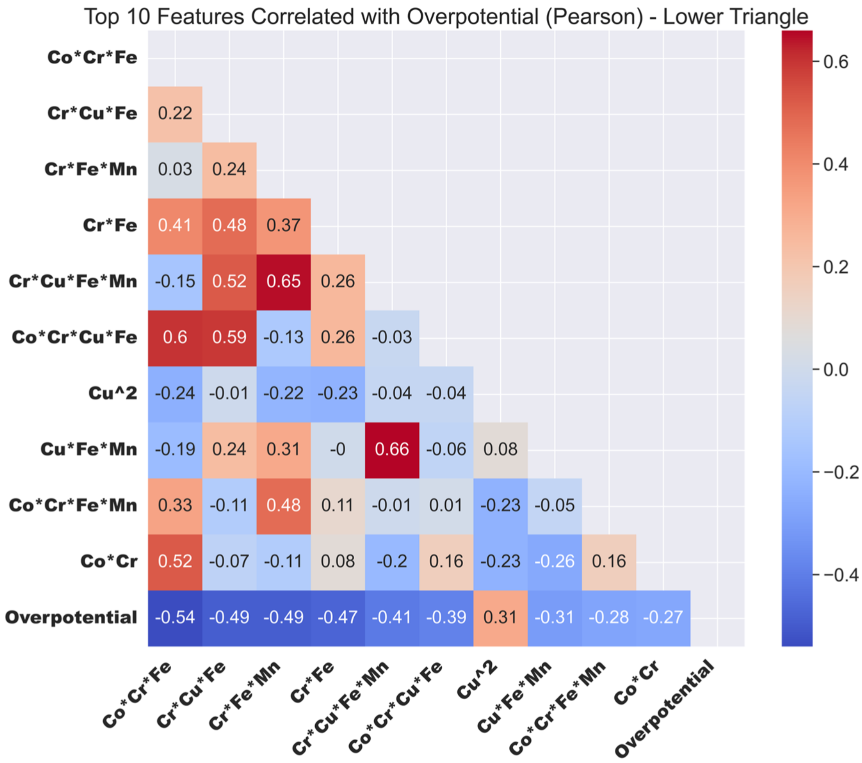

Figure 3. Correlation coefficients between overpotential and the top 10 features.

variables selected as features for the ML model included the elemental composition ratios of the five transition metals (Fe, Co, Cr, Mn, and Cu), while the target output was the measured overpotential. To generate the data set, a series of FCCMC LDHs catalysts with varying metal ratios were synthesized via a hydrothermal method under fixed reaction conditions (180 °C, 12 h). In total, 70 distinct compositions were prepared by systematically altering the relative molar ratios of the constituent elements within practical synthesis limits and corresponding over potentials, as outlined in Table S1 (Supporting Information). Each datum sample consists of the molar ratios of five elements: Fe, Co, Cr, Mn, and Cu. The molar ratio values range from 0 to 1, with the sum of all elements constrained to equal 1. All catalysts were deposited on nickel foam (NF) and their electrochemical performance was evaluated by linear sweep voltammetry (LSV) at a scan rate of 10 mV s^{-1} as shown in Figure S2 (Supporting Information). Based on LSV data, the overpotential required to achieve a current density of 10 mA cm^{-2} ranged from 286 mV to 408 mV, as illustrated in Figure S3 (Supporting Information). This experimental data set served as the initial training input for the machine learning models used to predict the optimal high-entropy LDHs formulation with enhanced OER activity.

Machine Learning-Guided Composition Optimization

Figure 2 presents an overview of the complete ML optimization workflow. To accelerate the discovery of high-performance electrocatalysts, a ML-assisted approach was adopted to optimize the composition of FeCoCrMnCu-based high-entropy LDHs for OER. A data set comprising 70 experimentally synthesized compositions was constructed, with corresponding overpotential values at 10 mA cm^{-2} used as the target property. Feature engineering was employed to generate interaction and polynomial terms from the elemental ratios, enabling the model to capture nonlinear relationships. The data were preprocessed and used to train an XGBoost regression model, which was subsequently optimized through cross-validation. This strategy enables the identification of an optimal composition with minimal overpotential, which is further validated experimentally.

The higher-order features in this study were generated by polynomial expansion of the original physicochemical features. The second to fifth-order interaction terms were constructed using a custom generate interactions function. As illustrated below, this function first identifies feature combinations using the combinations module and then generates all interaction terms up to the specified polynomial order. To avoid duplicated terms arising from feature order, all selected features were alphabetically sorted and named using the "*" delimiter. The numerical values of the interaction features were then obtained by performing row-wise multiplication and storing the results in the DataFrame.

The higher-order interaction terms were defined using a standard polynomial formulation. In matrix form, the expanded feature representation can be expressed as $\Phi(X)$ = [Fe, Co, Cr, Mn, Cu, Fe*Co, Fe*Cr···Fe*Co*Cr*Mn*Cu···Cu^5] where $\Phi(X)$ denotes the polynomial feature mapping commonly applied in machine learning models. The numerical values of these interaction terms were computed through element-wise multiplication of the corresponding base features. This procedure ensures that each generated interaction term directly reflects the mathematical product of its constituent variables.

To better capture the complex interactions among elements, this study employed feature engineering to generate high-order interaction terms (from second to fifth order, e.g., Fe*Co, Co*Cr, Fe*Mn*Cr) and square terms (e.g., Fe^2, Mn^2, Cr^2). These expanded features enhance the model's ability to learn nonlinear relationships. For instance, Fe*Co may reflect the synergistic behavior between iron and cobalt, while Cr^2 may indicate how variations in chromium content influence overpotential.

A preliminary assessment of feature relevance was performed using Pearson correlation analysis, defined as in eq 1

$$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (1)$$

where r ranges from −1 to 1, indicating the strength and direction of a linear relationship between two variables, that is, x_i and y_i. $\bar{x}_i$ and $\bar{y}_i$ are the mean of all x_i and y_i, respectively. Figure 3 presents the correlation matrix as a heatmap, highlighting the relationship between each feature and overpotential. Notably, the interaction term Co*Cr*Fe showed a correlation of −0.54 with overpotential, suggesting its

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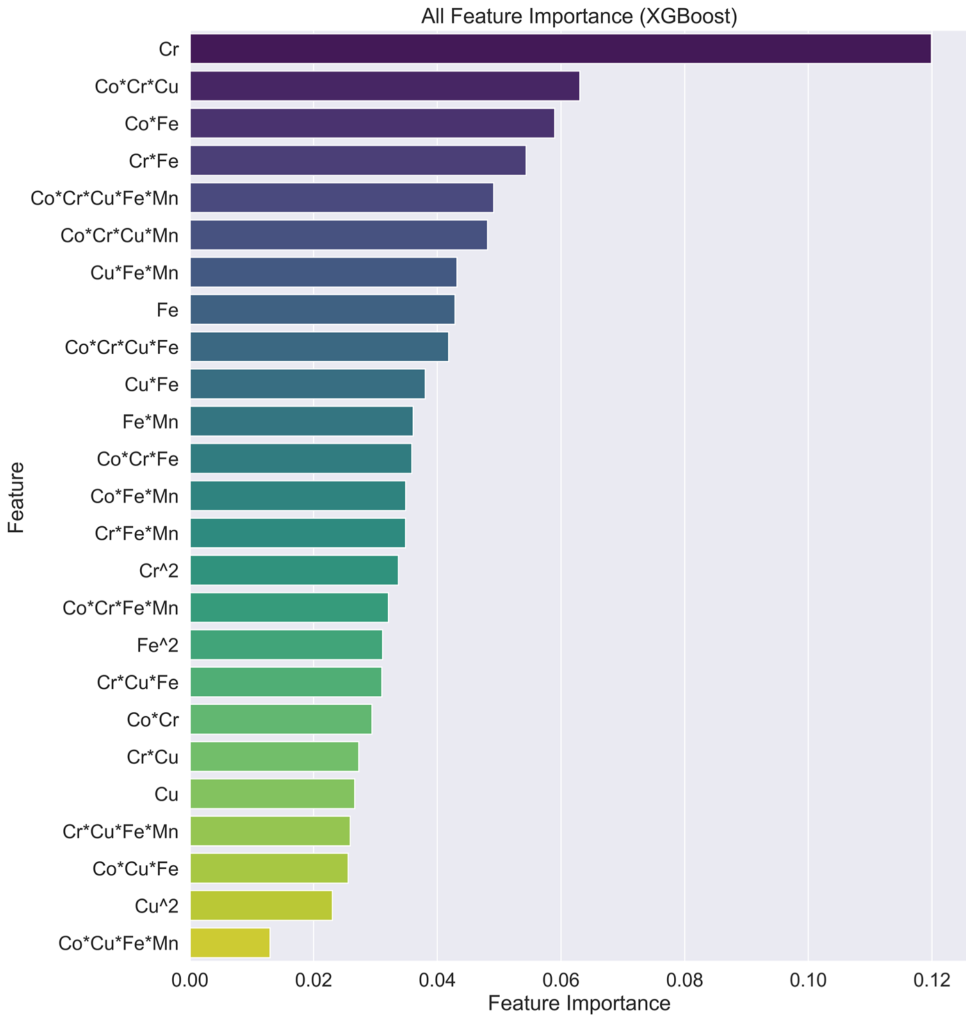

Figure 4. Bar chart of feature importance.

significant role in catalytic performance. Other features like Cr*Cu*Fe and Cr*Fe*Mn also exhibited strong correlations, emphasizing the predictive value of multielement interactions.

Model Selection and Training

Given the high dimensionality and potential nonlinear interactions in the FeCoCrMnCu high-entropy system, the Extreme Gradient Boosting (XGBoost) regression model was selected to predict OER overpotential. XGBoost is well-suited for the data set with complex feature spaces due to its robust regularization (L1 and L2), high efficiency, and strong interpretability compared to deep learning models. To capture higher-order interactions, second- to fifth-order polynomials and interaction terms were generated, significantly expanding the feature space. All features were standardized using the Yeo–Johnson transformation to normalize data distributions and improve model stability. To reduce redundancy and enhance performance, the SelectKBest method was applied for feature selection. The data set was split into training (80%) and testing (20%) subsets. A systematic workflow, implemented using Scikit-learn's pipeline, ensured consistent preprocessing, training, and evaluation, enabling reliable model generalization despite the data set's limited size. To reduce the risk of feature space inflation and multicollinearity, we utilized the SelectKBest feature selection method from scikit-learn. This module evaluates the linear relationship between each feature and the target variable (overpotential) using the f_regression metric, which conducts an F-test to derive *F*-scores. Further ranked all features by *F*-score and retained the top 25 most influential features for model training. The configuration was as follows: SelectKBest(f_regression), selection criterion: *F*-score, and the number of retained features: $K = 25$.

Model Evaluation and Hyperparameter Optimization

To evaluate the prediction accuracy and generalization of the XGBoost model, we used two standard metrics: root mean squared error (RMSE) and the coefficient of determination (R^2). RMSE measures the average deviation between predicted and actual overpotentials, while R^2 assesses the proportion of variance explained by the model, with values closer to 1 indicating better performance. RMSE (eq 2) provides a direct indication of prediction accuracy. However, RMSE is sensitive to the scale of the data and does not reflect how well the model captures variance. In a data set with a wide range, a large RMSE does not necessarily imply poor performance. To address this limitation, R^2 (eq 3) is also used as an evaluation metric. R^2 ranges from 0 to 1, with values closer to 1 indicating a stronger ability to explain variance and a better model fit. An R^2 near 0 suggests the model performs no better than predicting the mean.

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(\widehat{\eta_i} - \eta_i)^2} \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(\widehat{\eta_i} - \eta_i)^2}{\sum_{i=1}^{m}(\overline{\eta_i} - \eta_i)^2} \tag{3}$$

here, η_i is the actual overpotential, $\widehat{\eta_i}$ is the predicted value, $\overline{\eta_i}$ is the mean of the actual overpotentials, and m is the total number of samples.

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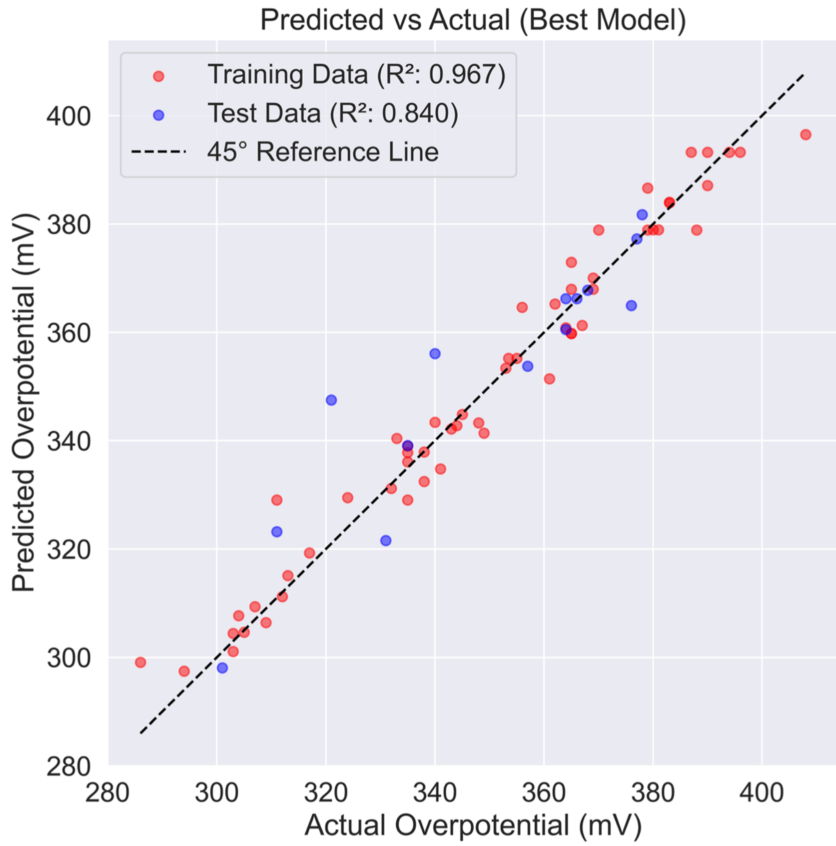

Figure 5. Scatter plot of predicted vs actual values.

Model performance was optimized through grid search, focusing on hyperparameters such as the number of selected features, feature and sample subsampling ratios, learning rate, maximum tree depth, minimum child weight, number of trees, and regularization terms (L1, L2). The optimal parameters were:

- Feature selection = 25
- Feature subsampling ratio = 0.3
- Learning rate = 0.2
- Max tree depth = 1
- Min child weight = 5
- Number of trees = 1300
- Sample subsampling = 0.2
- L1 = 0.14, L2 = 0.8

These settings were chosen to balance model complexity and avoid overfitting, given the limited data set. Feature importance analysis using the gain metric revealed that Cr contributed most (~12%) to reducing overpotential, followed by Co*Cr*Cu (~6.4%) and Co*Fe (~5.9%), underscoring the dominant role of chromium and multielement interactions in enhancing performance (see Figure 4).

The results indicate that Cr and its interaction terms play a key role in overpotential prediction, highlighting that increasing the Cr content or optimizing Co*Fe synergy could potentially lower overpotential. The optimized model achieved high performance, with a training R^2 of 0.967 and a test R^2 of 0.84. To evaluate the model's robustness and generalization ability, 5-fold cross-validation was performed on the 56-sample training set. In each iteration, the data was split into 4 folds for training and 1 for validation, repeated five times. The average cross-validation results yielded an RMSE of 10.77 mV and an R^2 of 0.83. These are slightly lower than the training (RMSE: 5.41 mV, R^2: 0.967) but close to test set results (RMSE: 9.95 mV, R^2: 0.840), primarily due to the smaller training size in each fold. Nonetheless, the results demonstrate good model stability and generalization (see Figure 5). Residual plots (Figure S4) show that the prediction errors are centered around zero with no evident bias. Residuals follow a normal distribution for both training and test sets (Figure S5), while box plots of predicted values (Figure S6) display similar distributions across data sets, further confirming the model's reliability.

Prediction Results and Feature Importance Analysis

To improve the interpretability of the machine learning model and gain insights into the elemental contributions, SHapley Additive exPlanations (SHAP) analysis was employed. SHAP not only quantifies the overall importance of each feature but also reveals how specific values influence the predicted overpotential. As illustrated in Figure 6, the SHAP value distribution for the top 20 features indicates the relative impact of each feature. Notably, the interaction term Co*Fe exhibits the broadest SHAP range (approximately −16 mV to +14 mV), highlighting its strong influence. A higher Co*Fe value (shown in red) generally leads to a reduction in predicted overpotential, whereas lower values (blue) tend to increase it, confirming the strong synergistic role of Co*Fe interactions. The analysis confirms Co*Fe as the most influential feature with a generally negative effect on overpotential. However, the lowest overpotential is not always achieved at the highest Co*Fe value, suggesting a nonlinear relationship. Similarly, Cu*Fe and Cr*Fe interactions show wide SHAP distributions, further emphasizing their importance in overpotential prediction. Figure S7 displays the mean absolute SHAP values for the top 20 features, representing their average impact on predictions. Other impactful features include Cu*Fe, Cr*Fe, and higher-order terms like Co*Cu*Fe and Cu*Fe*Mn, highlighting the importance of multielement interactions.

In fact, the XGBoost gain metric and the SHAP analysis quantify feature influence at different conceptual levels. They provide complementary perspectives on the model's behavior. The gain metric in XGBoost is derived from the tree-splitting process and reflects each feature's contribution to reducing training error or loss during model construction. A higher gain for Cr indicates that this feature was frequently and effectively used as a splitting criterion

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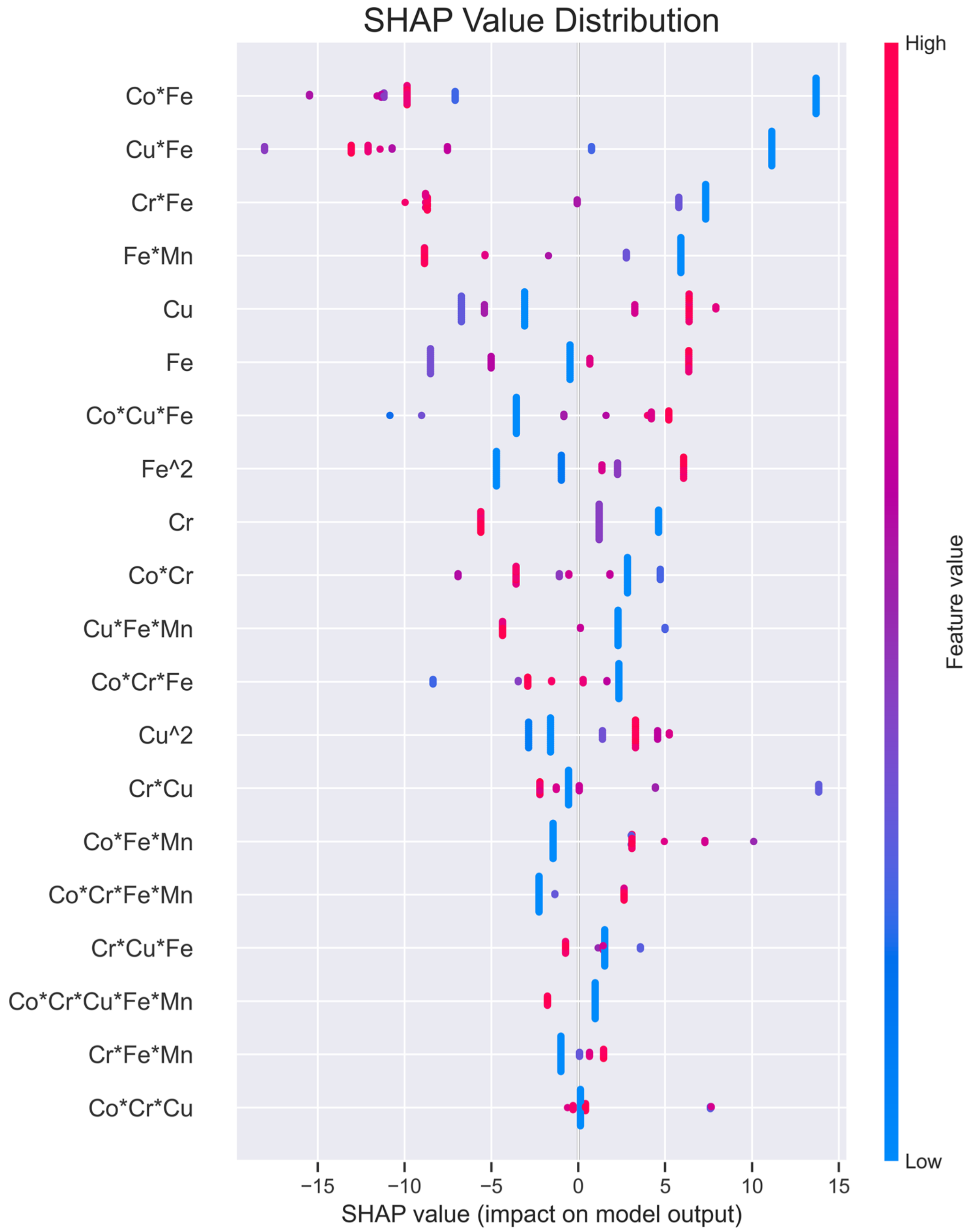

Figure 6. SHAP value distribution of top 20 features.

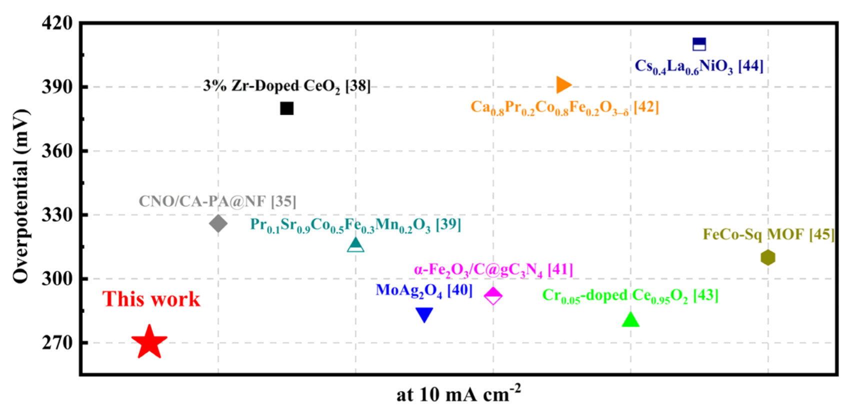

Figure 7. Comparison of the overpotential of the FeCoCrMnCu LDHs with those of recently reported electrocatalysts at 10 mA cm^{-2}.[35−45]

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across the boosted decision trees, meaning it is influential at the structural level of the model. In contrast, SHAP values are based on cooperative game theory and attribute each feature's marginal contribution to the final model output. SHAP explicitly accounts for feature interactions and provides both global and local interpretability. The finding that Co*Fe is the most influential feature in the SHAP analysis indicates that this interaction term contributes the most significant variation to the prediction-level output, even if it is not the most frequently used feature during tree splitting. In summary, the gain metric describes which features the model relies on most during training, whereas SHAP describes which features most strongly influence the final predictions.

To determine the optimal elemental composition, the trained XGBoost model was used to predict overpotentials for all possible molar ratio combinations of Fe, Co, Cr, Mn, and Cu (ranging from 0 to 1 in 0.05 increments, constrained to sum to 1). The composition with the lowest predicted overpotential was:

- Fe: 0.15, Co: 0.10, Cr: 0.30, Mn: 0.30, Cu: 0.15
- Predicted overpotential: 261.253 mV illustrates the distribution of predicted overpotentials, with the minimum marked by a red dashed line (Figure S8).

Figure 7 compares the overpotential of the optimized FeCoCrMn-Cu-LDHs with ML-designed OER catalysts, highlighting the strong performance of our high-entropy system and the effectiveness of machine learning-guided design.

Morphological and Chemical Evaluation of the Optimized Electrocatalyst

The crystalline structure of the ML-optimized FCCMC LDHs catalyst was examined using X-ray diffraction (XRD) analysis. As shown in Figure 8, the XRD pattern displays a series of sharp and well-defined

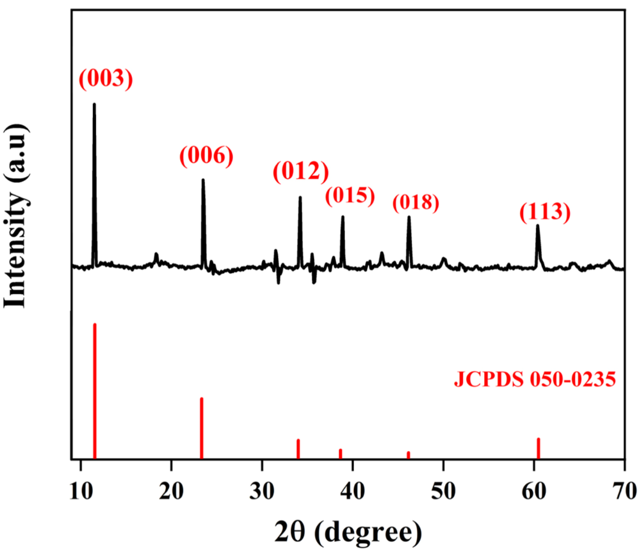

Figure 8. XRD spectrum of optimized FeCoCrMnCu LDHs.

diffraction peaks located at $2\theta \approx 11.6°$, 23.3°, 34.6°, 39.2°, 46.7°, and 60.6°, which are indexed to the (003), (006), (012), (015), (018), and (113) planes, respectively. These reflections are characteristic of LDHs phase with a hydrotalcite-like structure. The observed peak positions are in good agreement with the standard reference pattern for LDHs materials (JCPDS no. 050-0235), confirming the successful formation of a crystalline LDHs phase.[46] The intense and narrow (003) peak at low angle indicates a high degree of crystallinity and well-developed stacking order along the *c*-axis, which is typical for layered hydroxide materials. The absence of secondary phases or impurity peaks in the XRD profile suggests that all five metal cations (Fe, Co, Cr, Mn, Cu) were uniformly incorporated into a single-phase LDHs structure without phase segregation.

The surface morphology and nanostructure of the FCCMC LDHs were investigated using field emission scanning electron microscopy (FE-SEM) and field emission transmission electron microscopy (FE-TEM). As shown in Figure 9a−d, the FE-SEM images reveal that the catalyst exhibits a sheet-like morphology with a porous and interconnected architecture. At lower magnification (Figure 9a), the material appears uniformly deposited over the 3D scaffold of the nickel foam substrate, ensuring good electrical contact and mechanical integration.[47] Closer inspection at higher magnifications (Figure 9b−d) shows that the material is composed of thin, wrinkled nanosheets, loosely assembled to form a layered and porous network. These nanosheets appear to interlace and partially stack, forming voids and surface roughness that enhance electrolyte access and active site exposure. Such a morphology is typical of layered double hydroxides and is highly beneficial for electrocatalysis, as it maximizes the surface-to-volume ratio and facilitates ion diffusion and gas bubble release during oxygen evolution.[48]

Further insight into the structural features was obtained from FE-TEM, as shown in Figure 10a−c. The low-magnification TEM image (Figure 10a) displays transparent, sheet-like domains with irregular edges, indicative of ultrathin 2D layers. The high-magnification TEM image (Figure 10b) focusing on the edge of LDHs catalyst further confirms the stacked layered structure. The corresponding high-resolution TEM image (Figure 10c) clearly show well-resolved lattice fringes, confirming the crystalline nature of the material. The measured interplanar spacing of 0.248 nm corresponds to the (012) plane of LDHs-type structures, further verifying the successful formation of a layered phase. To further confirm the elemental distribution and homogeneity of the synthesized high-entropy FCCMC LDHs catalyst, energy-dispersive X-ray spectroscopy (EDX) elemental mapping was performed in conjunction with SEM analysis. The EDX mapping results, presented in the Supporting Information (Figure S9), clearly demonstrate the uniform spatial distribution of all five transition metals Fe, Co, Cr, Mn, and Cu throughout the LDHs matrix. These EDX results further validate the successful synthesis of a true high-entropy LDHs system as predicted by the machine learning model.

X-ray photoelectron spectroscopy (XPS) was also performed to probe the valence states of the transition metals present in the FeCoCrMnCu LDHs. The high-resolution spectra for Fe 2p, Co 2p, Cr 2p, Mn 2p, and Cu 2p regions are shown in Figure 11. The Fe 2p spectrum displays two main peaks around 710.6 eV (Fe $2p_{3/2}$) and 724.1 eV (Fe $2p_{1/2}$), accompanied by a prominent satellite feature (717.6 eV). These peaks are characteristic of Fe^{3+} in an octahedral environment, typical of LDHs frameworks. The satellite peak confirms the presence of Fe in a high-spin configuration.[49] The Co 2p region shows peaks at 781.0 eV (Co $2p_{3/2}$) and 796.0 eV (Co $2p_{1/2}$), along with broad satellite features around 786 and 803 eV. The main peak position and satellites suggest the coexistence of both Co^{2+} and Co^{3+}, implying mixed valence states which are known to enhance electron transfer during the OER process.[50,51] The Cr 2p region exhibits two prominent peaks at binding energies of approximately 577.1 and 586.6 eV, corresponding to Cr $2p_{3/2}$ and Cr $2p_{1/2}$, respectively. The Cr $2p_{3/2}$ peak can be deconvoluted into two components: one at 576.1 eV attributed to Cr^{2+}, and a smaller contribution at 577.4 eV indicating the presence of Cr^{3+}, confirming mixed valency.[52] The Mn 2p spectrum shows characteristic peaks at 641.5 and 653.1 eV, which are assigned to Mn $2p_{3/2}$ and Mn $2p_{1/2}$, respectively. The Mn $2p_{3/2}$ peak is further deconvoluted into two distinct components. The peak at 641.5 eV corresponds to Mn^{2+}, while the second peak at 643.4 eV is associated with Mn^{3+}. The coexistence of Mn^{3+} and Mn^{2+} indicates a multivalent state, which facilitates enhanced electron transfer kinetics during OER.[53] The broadness and asymmetry of the peaks suggest a mixed-valence state, which is beneficial for redox flexibility during OER. The Cu $2p_{3/2}$ peak appears around 933.0 eV, with a shakeup satellite around 941 eV, which confirms the presence of Cu^{2+}.[54] This oxidation state is known to promote oxygen intermediate stabilization and contributes to the multimetal synergy in high-entropy systems. As shown in Figure 11f, the high-resolution O 1s XPS spectrum of the HE-FCCMC LDH reveals three deconvoluted peaks centered at

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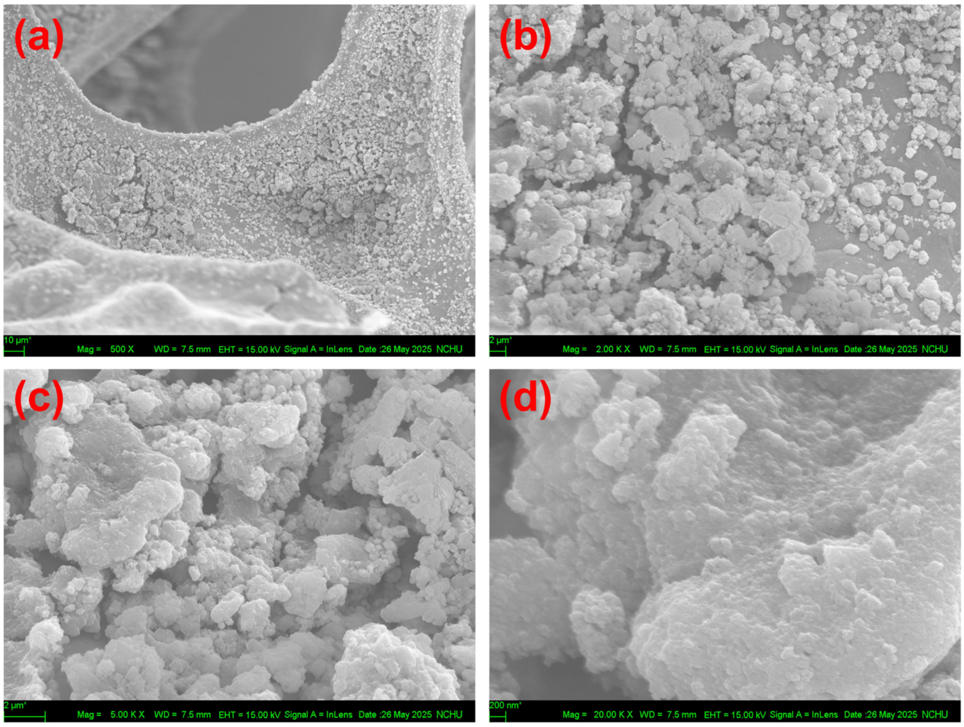

Figure 9. FE-SEM images of the optimized FeCoCrMnCu LDHs recorded at different magnifications: (a) 500×, (b) 2000×, (c) 5000×, and (d) 20,000×.

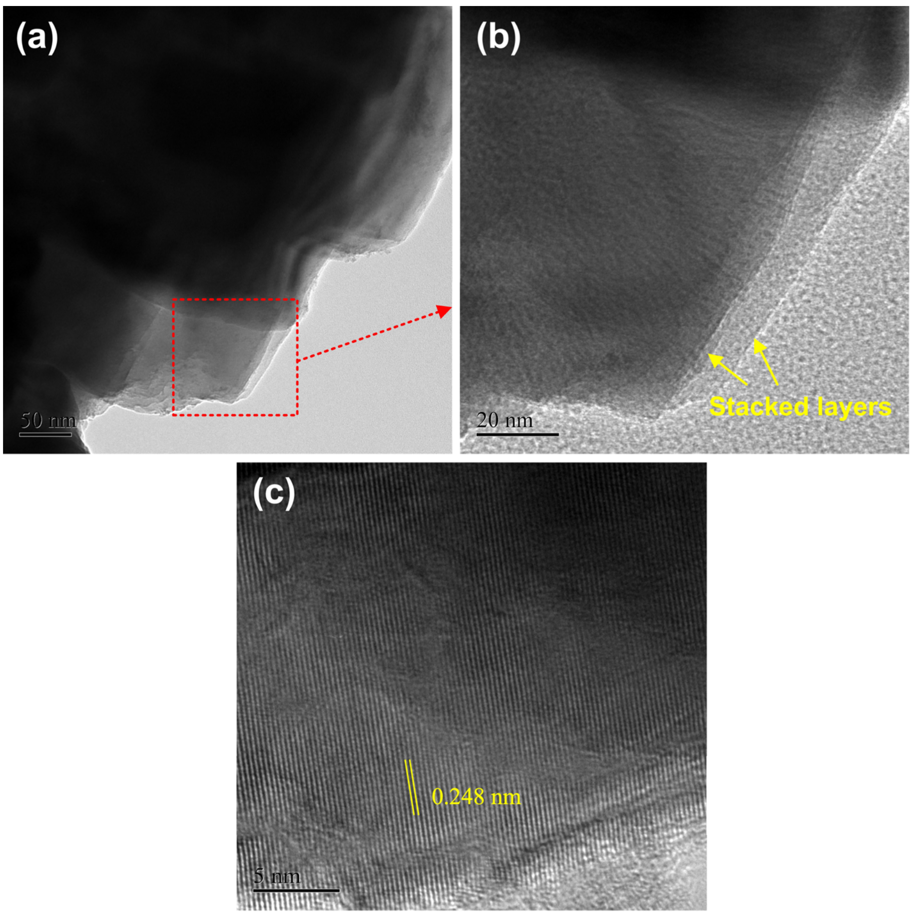

Figure 10. FE-TEM images of (a,b) low and high magnifications, and (c) lattice fringes of optimized FeCoCrMnCu LDHs.

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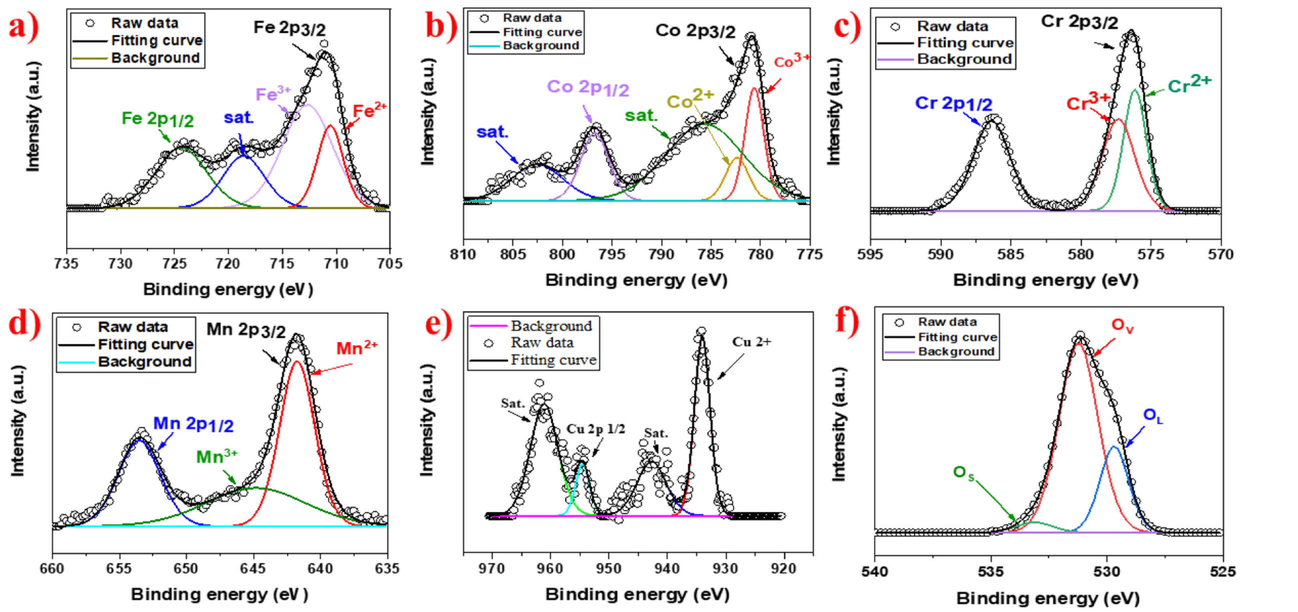

Figure 11. XPS spectra of (a) Fe 2p, (b) Co 2p, (c) Cr 2p, (d) Mn 2p, (e) Cu 2p, and (f) O 1s of optimized FeCoCrMnCu LDHs.

approximately 529.6, 531.5, and 532.8 eV. These peaks correspond to lattice oxygen (O_L), oxygen vacancies (O_V), and surface adsorbed oxygen species (O_S), respectively.[27] The peak at 529.6 eV is attributed to O_L, representing oxygen atoms incorporated within the crystalline metal hydroxide lattice. The dominant peak at 531.5 eV indicates a high concentration of O_V, associated with under-coordinated or defective oxygen species resulting from lattice distortion and nonstoichiometry. The third peak at 532.8 eV corresponds to O_S, including surface hydroxyls, carbonates, or molecularly adsorbed water. Quantitative analysis shows that O_V constitutes 72% of the total oxygen content, while O_L and O_S contribute 24% and 4%, respectively. The high proportion of oxygen vacancies is noteworthy, as it enhances electrical conductivity by increasing carrier mobility, accelerates OER kinetics through improved OH^- adsorption and intermediate formation (*O, *OOH), and promotes strong metal oxygen hybridization, thereby optimizing the electronic environment of active sites such as Fe, Co, Mn, and Cu.[55] This clearly confirms that the OER follows the lattice oxygen mechanism (LOM). These XPS results corroborate the electrochemical findings, indicating that the rich defect chemistry and high surface activity of the FeCoCrMnCu LDHs contribute significantly to its superior OER performance.

The textural properties of the ML-optimized FCCMC LDHs catalyst were investigated using nitrogen adsorption−desorption isotherms (Figure 12). The analysis yielded a Brunauer−Emmett−Teller (BET) surface area of 233.5 $m^2\ g^{-1}$, indicating a highly porous architecture with extensive surface exposure. Such a large surface area is a key factor in enhancing electrocatalytic performance, particularly in multistep reactions like the OER, which require efficient adsorption, activation, and desorption of reaction intermediates. A high surface area provides a greater density of electrochemically accessible active sites, allowing for more catalytic events to occur simultaneously. It also promotes enhanced interaction with the electrolyte, improving ion transport and facilitating faster reaction kinetics.[56] Additionally, the interconnected porous network supports efficient diffusion of reactants and release of evolved O_2 gas, minimizing mass transport limitations during prolonged electrolysis.

To confirm the elemental composition of the synthesized FCCMC LDHs catalyst, inductively coupled plasma optical emission spectrometry (ICP−OES) analysis was conducted (Figure 13). The results verified the successful incorporation of all five target transition metals in proportions closely matching the machine learning-optimized design. The elemental ratios were determined to be Fe (14.9%), Co (10.3%), Cr (29.5%), Mn (29.0%), and Cu (16.3%), which align well with the predicted composition of

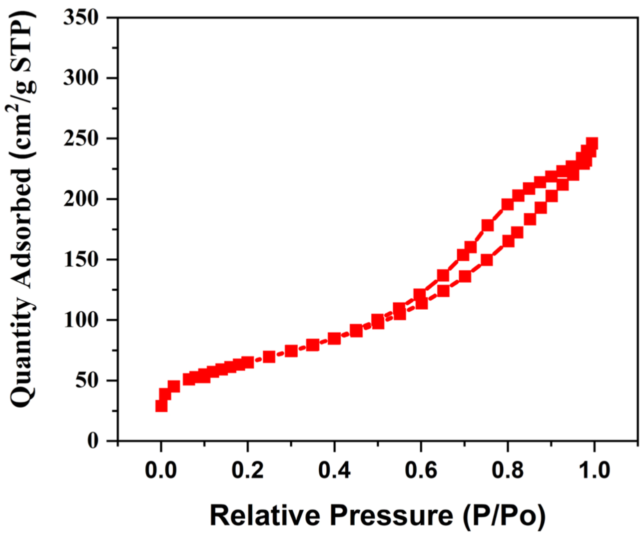

Figure 12. BET isotherm of optimized FeCoCrMnCu LDHs.

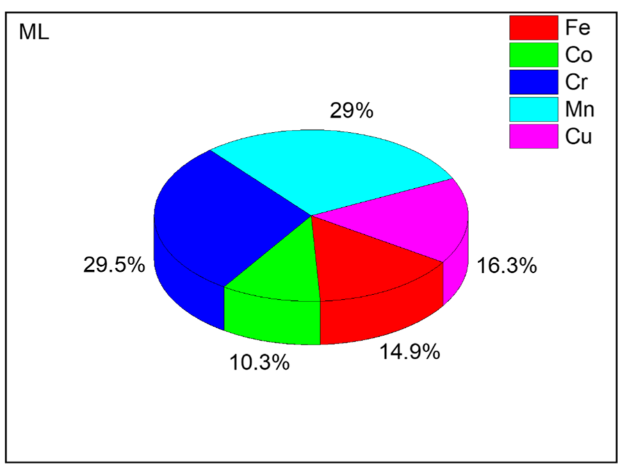

Figure 13. ICP−OES data of optimized FeCoCrMnCu LDHs.

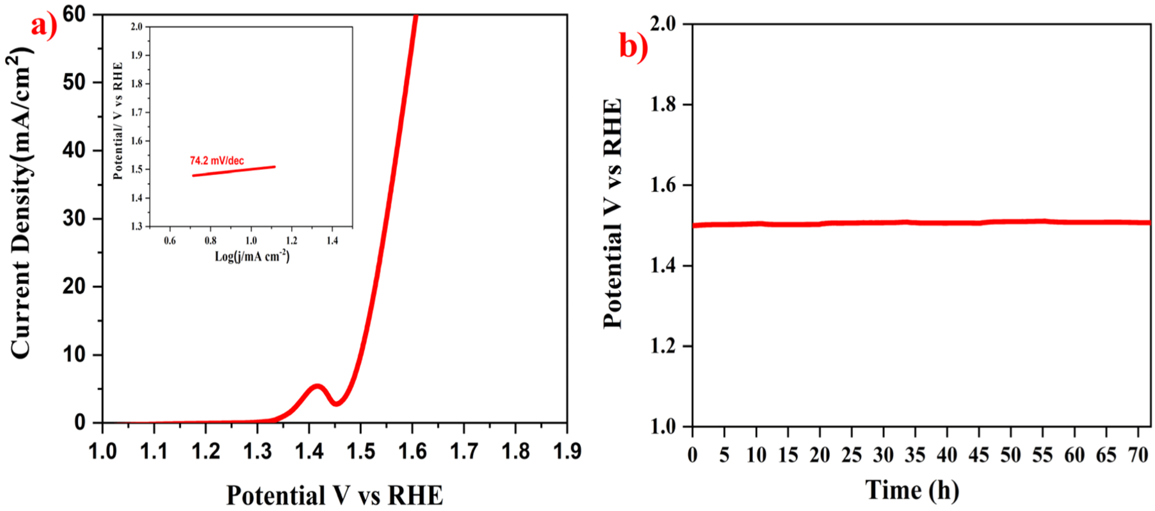

Figure 14. (a) LSV curve and (b) chronopotentiometry analysis of FeCoCrMnCu LDHs (inside in a) Tafel slope.

$Fe_{0.15}Co_{0.10}Cr_{0.30}Mn_{0.30}Cu_{0.15}$. This close agreement between the experimental ICP−OES results and the ML-predicted composition confirms the accuracy and reliability of the machine learning-assisted optimization strategy. The ability to precisely synthesize the targeted high-entropy formulation is critical for validating the model's predictions and achieving the desired catalytic properties.

Electrocatalytic OER Performance

The OER performance of the ML-optimized FCCMC LDHs catalyst was systematically investigated in 1.0 M KOH using a conventional three-electrode setup, consisting of nickel foam as the working electrode, a platinum wire as the counter electrode, and a Hg/HgO electrode as the reference. The catalyst ink was uniformly coated onto the pretreated nickel foam to ensure reliable contact and minimize resistance effects. As shown in Figure 14a, linear sweep voltammetry (LSV) revealed that the catalyst achieved a current density of 10 mA cm^{-2} at overpotential of 270 mV. This low overpotential reflects the excellent electrocatalytic activity of the FCCMC LDHs and demonstrates the effectiveness of machine learning-guided composition optimization in designing high-performance electrocatalysts. Additionally, the Tafel slope was determined to be 74.2 mV dec^{-1} (the inset in Figure 14a), indicating favorable OER kinetics and a fast charge transfer process at the electrode−electrolyte interface. These results suggest that the catalyst can efficiently facilitate the multistep electron−proton transfer involved in the oxygen evolution reaction.[57]

To evaluate the long-term durability of the ML-optimized FCCMC LDHs electrocatalyst under operational conditions, chronopotentiometry measurements were performed in 1.0 M KOH at a constant current density of 10 mA cm^{-2} over an extended period of 72 h. As shown in Figure 14b, the potential remained remarkably stable throughout the duration of the test, with only negligible voltage drift observed. This indicates excellent structural and electrochemical stability of the catalyst under continuous alkaline OER operation.

The electrochemically active surface area (ECSA) of the ML-optimized FCCMC LDHs was estimated from the double-layer capacitance (C_{dl}) extracted from CV measurements in the nonfaradaic region (10−100 mV s^{-1}). A linear fit of the capacitive current versus scan rate (Figure S10a,b) yielded a C_{dl} of 4.3 μF cm^{-2}, corresponding to an ECSA of 0.1075 cm^2 (eq 4). This elevated ECSA reflects a high density of accessible active sites that can promote efficient charge transfer during OER.[58]

$$\mathrm{ECSA} = C_{dl}/C_s \tag{4}$$

To evaluate the intrinsic catalytic activity independent of morphology, the turnover frequency (TOF) was calculated at an overpotential of 400 mV using the redox-active surface area (Figure S10c,d). The FCCMC catalyst delivered a TOF of 0.225 s^{-1}, confirming its strong intrinsic OER activity.[59] The calculation methodology and relevant equations are provided in our previous work.[33]

Compared to conventional trial-and-error optimization, which would require the preparation and testing of more than 10,626 possible compositions, our ML-guided approach achieved reliable predictions using only 70 experimental data points. To ensure the model is used within its valid scope when identifying optimal material compositions, we restricted all virtual compositions to the model's interpretable domain, thereby avoiding unreliable extrapolations. Specifically, we implemented a feature range constraint that limits the elemental feature values of all virtual compositions to the minimum and maximum ranges observed across the 70 training samples, which guarantees that all prediction points remain within the data distribution's interpolation region rather than falling outside the training domain, thereby ensuring a reasonable level of reliability. This represents a remarkable 99.3% reduction in the time and effort required for catalyst discovery, while still delivering accurate predictions of the optimal high-entropy FeCoCrMnCu-LDHs composition (Figure 15). Such efficiency highlights the transformative role of ML in accelerating the design of next-generation electrocatalysts.

To ensure a reliable assessment of model performance and generalization capability, *k*-fold cross-validation was employed rather

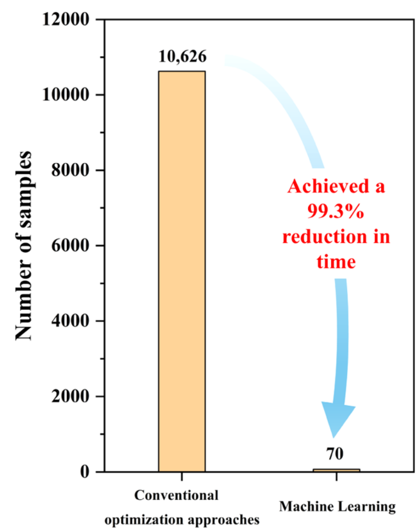

Figure 15. Bar chart for the comparison of conventional and ML approaches.

than relying on a single random train test split. A one-time split can be sensitive to sampling bias, particularly for limited data sets, and may lead to over or underestimation of predictive performance. In contrast, k-fold cross-validation provides a more robust and statistically meaningful evaluation by systematically rotating the test set across the entire data set. In this approach, the data set is divided into k equally sized subsets; in each iteration, one subset is used exclusively for testing while the remaining subsets are used for training, ensuring that every data point serves as a test sample exactly once. In this study, 5-fold cross-validation was adopted, with 14 samples used as the test set in each fold, enabling a comprehensive evaluation of model stability and predictive reliability despite the limited data set size. The reported prediction error of approximately 3% corresponds to the machine learning-identified optimal composition selected from the 10,626 possible elemental combinations. The model-predicted overpotential for this composition was directly compared with the experimentally measured value, demonstrating close agreement and validating the predictive accuracy of the trained model. To further assess robustness beyond cross-validation, three additional compositions that were not included in either the training or validation folds were independently synthesized and experimentally evaluated (see Table S2). The consistent agreement between predicted and experimental trends for these new samples provides additional evidence of the model's generalization capability and predictive reliability.

■ CONCLUSIONS

In summary, we developed a machine learning-guided framework for the targeted design of high-entropy FeCoCrMnCu layered double hydroxides (LDHs) electrocatalyst toward efficient oxygen evolution in alkaline media. The XGBoost regression model, trained on 70 experimentally measured compositions, exhibited strong predictive power (R^2 = 0.84), and successfully identified an optimal composition ($Fe_{0.15}Co_{0.10}Cr_{0.30}Mn_{0.30}Cu_{0.15}$) with a predicted overpotential of 261.253 mV. The experimentally synthesized catalyst showed a close agreement, achieving an overpotential of 270 mV at 10 mA cm^{-2} with only ~3% deviation, validating the ML approach. Structural characterization confirmed the formation of a single-phase LDHs with a porous nanosheet morphology that enhances active site accessibility. ICP−OES analysis verified the elemental composition precisely matched the ML predicted ratios. Electrochemical characterization revealed a favorable Tafel slope of 74.2 mV dec^{-1} and an excellent turnover frequency of 0.225 s^{-1}, highlighting its intrinsic catalytic activity. This work underscores the power of combining machine learning and entropy-driven design for accelerating the discovery of next-generation OER electrocatalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c07303.

Table of initial 70 experimental data set used for ML training; additional data sets of FeCoCrMnCu-LDHs compositions; histogram of Fe, Co, Cr, Mn and Cu distribution in the initial data set; LSV curves of 70 different high entropy FeCoCrMnCu LDHs; histogram of overpotential distribution in the initial data set; residual plot; distribution plot of prediction errors; Box plot of predicted values; mean absolute SHAP values; distribution plot of predicted overpotential; bright field images; EDX mappings of Fe, Co, Cr, Mn, and Cu elements of optimized FeCoCrMnCu LDHs: ECSA and TOF studies of optimized FeCoCrMnCu LDHs; link for experimental data set (PDF)

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Author Contributions

[#]C.P. and C.-F.H. contributed equally to this work. Chandrasekaran Pitchai: methodology, validation, formal analysis, visualization, investigation, writing—original draft. Chao-Fang Huang: methodology, software, validation, formal analysis, investigation, data curation, visualization, writing—original draft. Ting-Yu Lo: software, validation, formal analysis. Hung-Chung Li: conceptualization, software, supervision, data curation, writing—review and editing. Ming-Der Yang: conceptualization, supervision, funding acquisition, writing—review and editing. Chih-Ming Chen: conceptualization, investigation, resources, supervision, funding acquisition, writing—review and editing.

Notes

The authors declare no competing financial interest.

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